

PDF → Chapter 7 Outline

Active Scanning
Passive Scanning

Elements of a Wireless Network

① Wireless Network Infrastructure

② Wireless Hosts

↳ Laptop, Smartphone, IoT

↳ Run applications

↳ Stationary (non-mobile) or mobile

Wireless does not always mean mobility.

③ Base Stations

↳ Typically connected to wired network

↳ Relay- Responsible for sending packets ...

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④ Wireless Link

↳ Typically used to connect mobiles to base stations

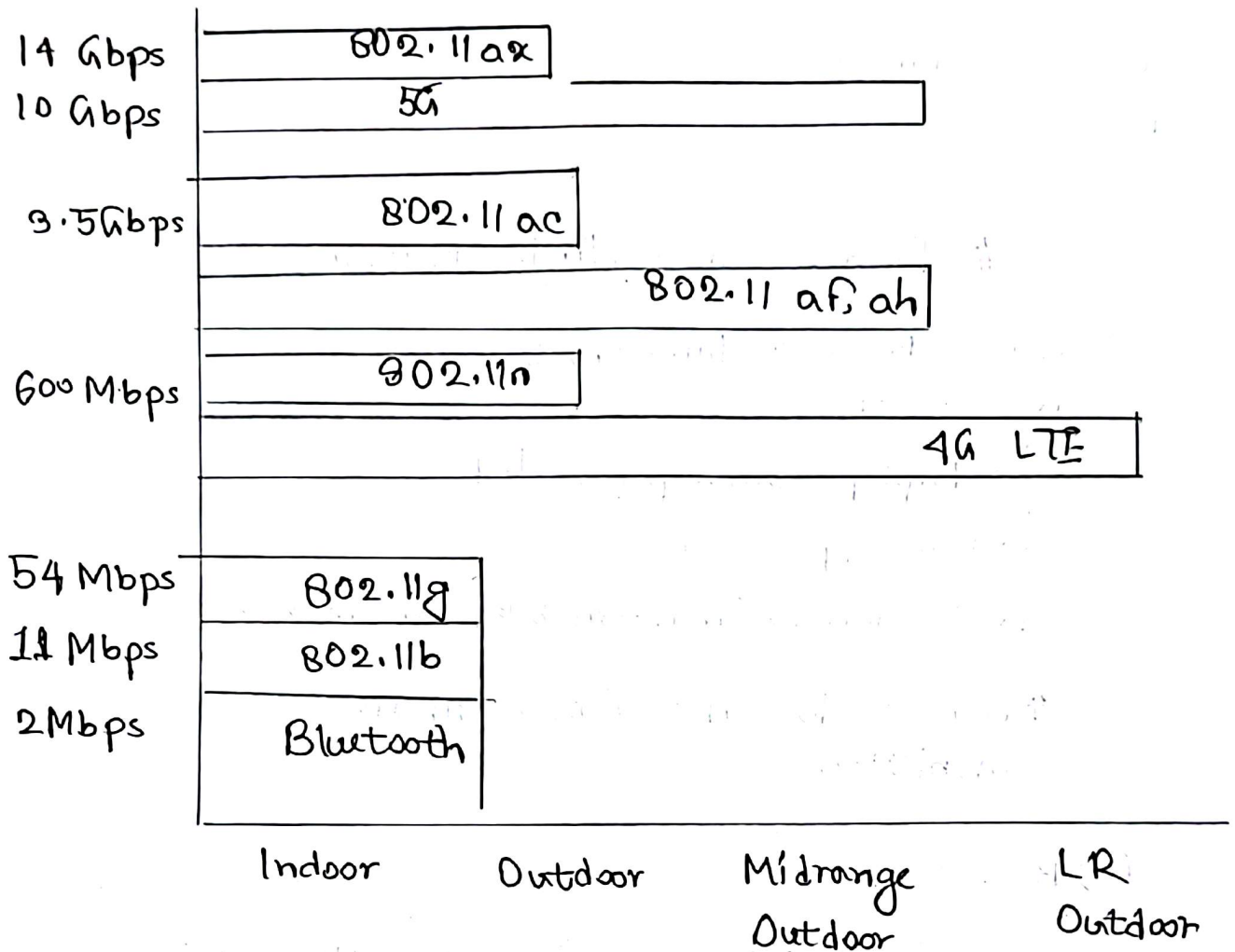
infrastructure mode

↳ Base Station connects mobile into wired network

↳ Handoff: Mobile Change ... PDF

Characteristics of Selected Wireless Links

Shayan



ad hoc mode

- No base stations
- Nodes can only transmit to other nodes within link coverage
- Nodes organize themselves into a network
↳ route among themselves

MANET
FANET
WANET
VANET

No base stations
or cell towers

Wireless Network Taxonomy

Shayan

	Single hop	multiple hop
infrastructure	host $\xrightarrow{\text{connects}}$ base station BS = Wifi, cellular $\rightarrow$ larger network Internet	hosts relay through several wireless nodes to connect to larger net (Mesh)
no-infrastructure	no base station, no connection to larger Internet (BL, AD HOC)	no base station, no connection to larger Internet. May relay like VANET

Wireless Link Characteristics

Importance differences from wired link.

- ☒ Decreased signal strength $\frac{1}{d^3} / \frac{1}{d^4}$
- ☒ Interference from other sources (Wireless, Cellular, Zigby)
- ☒ Multipath propagation

... make communication across (even a point-to-point) wireless ~~network~~ link much more difficult.

SNR = signal to Noise Ratio

larger SNR $\rightarrow$ easier to extract signal from noise
(this is good)

SNR versus BER tradeoffs $\rightarrow$ [A] PDF

SNR may change with mobility

Problems

$\hookrightarrow$ Hidden Terminal Problem

$\hookrightarrow$ Exposed Terminal Problem

$\uparrow$ self study

Signal Attenuation

CDMA $\leftarrow$ লাগবে না

IEEE 802.11 Wireless LAN $\leftarrow$ লাগবে

[A] PDF থেকে Table

* all use CSMA/CA for multiple access & have base stations and ad hoc network versions

802.11 LAN Architecture